Clinical Study Summary Report (CSSR) Document Status: Final | **Version:** 1.0 | **Date:** 03-Apr-2026

---

1. Administrative & Study Identification

Full Study Title: Phase I/IIa Dose Escalation Safety and Efficacy Study of Human Embryonic Stem Cell-Derived Retinal Pigment Epithelium Cells Transplanted Subretinally in Patients With Advanced Dry-Form Age-Related Macular Degeneration (Geographic Atrophy) **Short Title/Acronym:** OpRegen Phase 1/2a Safety and Efficacy Study **Protocol Number:** GR44172 **NCT Number:** NCT02286089 **Sponsor Name:** Lineage Cell Therapeutics / Roche / Genentech **Investigational Product:** OpRegen (RG6501) - human embryonic stem cell-derived retinal pigment epithelial (RPE) cells **Phase of Development:** Phase I/IIa **Medical Monitor:** Sponsor Clinical Operations Lead / Medical Monitor

---

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of subretinal transplantation of OpRegen (human embryonic stem cell-derived RPE cells) in patients with progressive dry age-related macular degeneration (AMD). **Secondary Objectives:** To explore the preliminary efficacy of transplanted OpRegen cells, specifically their ability to engraft, survive, moderate geographic atrophy (GA) disease progression, and improve Best Corrected Visual Acuity (BCVA) and retinal structure.

2.2 Background & Rationale In AMD, the accumulation of waste products (such as lipofuscin) in the retinal pigment epithelium leads to the dysfunction and death of RPE cells, ultimately driving the development of geographic atrophy (GA). There are limited treatment options to reverse this damage. Subretinal delivery of OpRegen allogeneic RPE cell therapy has the potential to counteract RPE cell loss by replacing missing cells, supporting overall retinal cell health, and improving both the structure and function of the retina.

---

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A (Dose-escalation across 4 sequential cohorts)

3.2 Planned Enrollment & Duration Target N: Approximately 24 subjects **Number of Sites:** Multiple centers (including sites in the United States and Israel) **Estimated Study Start:** 31-Mar-2015 **Estimated Primary Completion:** Dec-2020 (with long-term follow-up extending to 2031)

---

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 50 years. 2. Confirmed diagnosis of non-neovascular (dry) AMD with funduscopic findings of geographic atrophy (GA) in the macula of both eyes. 3. Best Corrected Visual Acuity (BCVA) of 20/200 or worse in the study eye (criteria adapted to $\le 20/100$ in subsequent cohorts upon established safety). 4. Ability to undergo a vitreoretinal surgical procedure under monitored anesthesia care.

4.2 Exclusion Criteria 1. History of retinal detachment repair or prior cataract removal within 2 months in the study eye. 2. History of any condition other than AMD associated with choroidal neovascularization. 3. Positive screening for active systemic infections (HIV, Hepatitis B/C, CMV IgM, Epstein-Barr Virus IgM, Tuberculosis). 4. Contraindication for receiving systemic immunosuppression.

---

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single administration of OpRegen (RG6501) cell suspension via dose-escalation. **Route of Administration:** Subretinal delivery (surgical vitreoretinal implantation). **Duration of Treatment:** Single intervention with long-term follow-up lasting up to 36 months and beyond.

5.2 Concomitant & Prohibited Therapy Permitted: A short course of systemic immunosuppressant medications is required to prevent an immune reaction to the allogeneic hESC-derived RPE cells. **Prohibited:** Current participation in another clinical study or concurrent investigational treatments.

---

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, BCVA baseline, OCT Imaging, Viral & Infection Labs
Baseline	Day 0	Vitreoretinal surgery, OpRegen Subretinal Implantation, Immunosuppression
Interim	Months 1 to 24	Safety & engraftment assessment, BCVA testing, OCT structure analysis
End of Study	Month 36+	Long-term follow-up evaluation for retinal structural preservation and visual acuity gains

---

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence, frequency, and severity of treatment-emergent Adverse Events (AEs) to evaluate overall safety and tolerability of the OpRegen procedure. **Measurement Method:** Clinical monitoring and surgical safety evaluation.

7.2 Secondary & Exploratory Endpoints 1. Change from baseline in Best Corrected Visual Acuity (BCVA), measured by ETDRS letters. 2. Evidence of retinal structural improvement, survival of transplanted cells, and moderation of geographic atrophy progression as assessed by quantitative optical coherence tomography (OCT).

---

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Extensive tracking of post-surgical complications, ocular inflammation, and potential immune rejection. **Serious Adverse Events (SAEs):** Reported promptly following standard clinical trial monitoring timelines. **Data Monitoring Committee (DMC):** External independent safety reviews were conducted between dose-escalating cohorts to approve the transition to less severely impaired patients.

---

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set consisting of all participants who received the subretinal injection. **Sample Size Justification:** $N=24$ divided into four cohorts provides adequate powering to assess primary safety and dose tolerability in a Phase I/IIa setting. **Handling of Missing Data:** Standard handling for long-term patient withdrawal, using available longitudinal OCT and BCVA records for completed months.

---

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study is conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Strict clinical oversight of the subretinal injection protocol and longitudinal site evaluation. **Audit Trail:** Secure EDC database for clinical assessments, centralized reading centers for objective, unbiased interpretation of OCT imaging.

---

11. Study Identifiers & Governance

Primary ID: GR44172 **Registry Number:** NCT02286089 **Sponsor/Lead Organization:** Lineage Cell Therapeutics / Roche / Genentech **Study Title:** Safety and Efficacy Study of OpRegen for Treatment of Advanced Dry-Form Age-Related Macular Degeneration **Official Regulatory Title:** Phase I/IIa Dose Escalation Safety and Efficacy Study of Human Embryonic Stem Cell-Derived Retinal Pigment

Epithelium Cells Transplanted Subretinally in Patients With Advanced Dry-Form Age-Related Macular Degeneration (Geographic Atrophy)

---

12. Clinical Framework (Study Specific)

Primary Condition: Geographic Atrophy (GA) secondary to Age-Related Macular Degeneration (AMD) **Diagnosis Threshold:** Funduscopic findings of GA, BCVA 20/200 or worse **Medical Methodology:** Subretinal transplantation of allogeneic RPE cells **Optimization Target:** Counteract RPE cell loss and improve visual acuity / retinal structure

---

13. Study Procedures & Methodology

Management Pathway: Vitreoretinal surgery for targeted subretinal cell delivery **Education Protocol (HCP):** Specialized surgical training for subretinal delivery and handling of the OpRegen (RG6501) cell suspension **Education Protocol (Patient):** Post-operative surgical care and strict adherence to the systemic immunosuppression protocol **Quality Audit Mechanism:** Centralized, quantitative Optical Coherence Tomography (OCT) reading and standard BCVA visual assessments

---

14. Observation & Follow-Up Schedule

Total Duration: 36 months (with extensions for ongoing long-term monitoring) **Onsite Visit Cadence:** Post-operative intervals culminating in long-term evaluations at Months 12, 24, and 36 **Remote Monitoring Frequency:** Intermittent protocol-defined safety/AE check-ins **Checklist Review Interval:** Regular clinical ophthalmic examinations per Phase 1/2a safety standards

---

15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	**[DD-MMM-YYYY]**
Clinical Operations Lead	**[Name]**	___	***[DD-MMM-YYYY]***
Lead Statistician	***[Name]***	_____	[DD-MMM-YYYY]